Polynomials and Factoring

variables: $x, y, z, a, b, c, r, s, t$, etc.

constants: $-2, 0, \pi, \sqrt[3]{5}, \frac{7}{2}$, etc.

A "term" is a product of constants and variables, usually written in a simplified form:

e.g. $(-3)\cdot\left(\frac{-1}{4}\right)\cdot x\cdot y\cdot x = \underline{\frac{3}{4}\cdot x^2\cdot y}$

A polynomial is a sum of terms.

1 term = "monomial"
2 " = "binomial"
3 " = "trinomial"
⋮

We will focus mostly on polynomials in only one variable.

j. $-x^3 + 3x^2 - 5x^1 - 7x^0$

$-1 \cdot x^3$

coefficients: $-1, 3, -5, -7$

leading term: $-x^3$

degree: 3

leading coefficient: -1

Constant polynomials are also included

Addition / Subtraction

$(x^3 + 2x^2 - 5x + 7) + (4x^3 - 5x^2 + 3)$

$= 5x^3 + (-3)x^2 - 5x + 10$

$= 5x^3 - 3x^2 - 5x + 10$

$$+2x^2-5x+7)-(4x^3-5x^2+3)$$

$$(x^3+2x^2-5x+7)-(4x^3-5x^2+3)$$

$$=(x^3+2x^2-5x+7)+(-4x^3+5x^2-3)$$

$$=-3x^3+7x^2-5x+4$$

Multiplication

$$(x^2+5x-4)(2x^3+3x-1)$$

$$=x^2(2x^3+3x-1)+5x(2x^3+3x-1)$$
$$-4(2x^3+3x-1)$$

$$=2x^5+3x^3-x^2+10x^4+15x^2-5x$$
$$-8x^3-12x+4$$

$$=2x^5+10x^4-5x^3+14x^2-17x+4$$

Factoring Identities

$x^2 - y^2 = (x+y)(x-y)$

$(x \pm y)^2 = x^2 \pm 2xy + y^2$

$(x \pm y)^3 = x^3 \pm 3x^2y + 3xy^2 \pm y^3$

$x^3 + y^3 = (x+y)(x^2 - xy + y^2)$

$x^3 - y^3 = (x-y)(x^2 + xy + y^2)$ $\quad 8a^3 = 2^3a^3 = (2a)^3$

e.g. $\cancel{8a^3 + b^3}$ $\quad 16a^3 + 2b^3 \qquad x = ?$

$2(8a^3 + b^3) \qquad y = ?$

$= 2(2a+b)(4a^2 - 2ab + b^2)$

e.g. $\cancel{x^4}$ $x^2 - 4y^2$

$$x^4 - 4y^2$$

$$\underbrace{x^4}_{(x^2)^2} \quad \underbrace{4y^2}_{(2y)^2}$$

$$= (x^2)^2 - (2y)^2$$

$$= (x^2 - 2y)(x^2 + 2y)$$

e.g. $4x^5y - 9x^3y^3$

$$= x^3(4x^2y - 9y^3)$$

$$= x^3y(4x^2 - 9y^2)$$

$$= x^3y[(2x)^2 - (3y)^2]$$

$$= x^3y[(2x - 3y)(2x + 3y)]$$

$$= x^3y(2x - 3y)(2x + 3y)$$

Factoring by Grouping

This can be thought of as a reverse of the distributive property / "FOIL"

$$ac + ad + bc + bd = a(c+d) + b(c+d)$$
$$= (a+b)(c+d)$$

e.g. $3x^3 + 2x^2 - 12x - 8$

$$= x^2(3x+2) - 4(3x+2)$$

$$= (x^2 - 4)(3x+2)$$

$$= (x-2)(x+2)(3x+2)$$